

§2.1

Addition and Subtraction Problem Types

Four structures, and how children make sense of them

Today's Goal

After completing this section, you should be able to:

- Differentiate between the different types of addition and subtraction problems based on context
- Understand how the different types of addition and subtraction problems influence the models and strategies used

LT.3 Addition and Subtraction Problem Types

I can identify by name and create different types of story problems (e.g., Take From Result Unknown) related to addition and subtraction.

How Old Is the Shepherd?

**There are 125 sheep and
5 dogs in a flock. How
old is the shepherd?**

▶ **Play: students answer this question**

youtu.be/kibaFBgaPx4

Why do so many children give a numerical answer? What does it mean to **make sense** of a problem?

One Fact, Four Problems

1. Eliz has six cookies. She ate two of them. How many cookies does Eliz have left?
2. Eliz has two dollars to buy cookies. How many more dollars does she need to earn to have six dollars?
3. Eliz has two dollars. Together with Tom, they have a total of six dollars. How many dollars does Tom have?
4. Eliz has two dollars. Tom has six dollars. How many more dollars does Tom have than Eliz?

Each appears to use the same number fact, $6 - 2 = 4$, but the **structures** are different.

Four Types, Three Subtypes Each

Table 2.1.1: Addition and Subtraction Problem Types

Type	Subtypes		
Add To	Result Unknown	Change Unknown	Start Unknown
Take From	Result Unknown	Change Unknown	Start Unknown
Put Together / Take Apart	Total / Whole Unknown	Addend / Part Unknown	Both Addends / Both Parts Unknown
Additive Compare	Difference Unknown	Bigger Unknown	Smaller Unknown

Add To Problems

Add To problems are also sometimes referred to as Join problems.

Definition 2.1.2

An **Add To problem** involves an action that increases the size of a set by a particular amount.

The action **increases** the size of the set, so the number fact is

$$\underline{\text{start}} + \underline{\text{change}} = \underline{\text{result}}$$

The Three Add To Subtypes

Table 2.1.3: Add To Problem Subtypes

Add To (Result Unknown)	Add To (Change Unknown)	Add To (Start Unknown)
Three people are in a line. Five more people get in the line. How many people are in the line now?	Three people are in a line. Some more people get in the line. Now there are eight people in the line. How many more people got in the line?	Some people are in a line. Five more people get in the line. Now there are eight people in the line. How many people were in the line before?
$3 + 5 = ?$	$3 + ? = 8$	$? + 5 = 8$

How Do Children Solve These?

One approach a child may use is called **direct modeling**.

Definition 2.1.5

Direct modeling is an approach to problem solving in which the solution to the problem is obtained by constructing a model of the problem. The model may use physical objects or may be drawn out.

Today you will do the same: model each problem with the cubes at your table.

The student videos we will watch are from *Children's Mathematics* (Cognitively Guided Instruction, Thomas Carpenter and colleagues).

The Toy Cars Problem

Alexis has some toy cars. He went to the store and bought 4 more toy cars and then he had 9 toy cars. How many toy cars did Alexis have to start with?

- a. **Face Partner B:** Use the cubes at your table to show your face partner how a child may use direct modeling to solve this problem.
Face Partner A: How else might a child go about solving this problem in a way that demonstrates making sense of the problem?
- b.  **Play: a student solves this problem** smarturl.it/cm3.6
- c. What type (and subtype) of addition or subtraction problem is this an example of?

Take From Problems

Take From problems are also sometimes referred to as Separate problems.

Definition 2.1.8

A **Take From problem** involves an action that decreases the size of a set by a particular amount.

The action **decreases** the size of the set, so the number fact is

$$\underline{\text{start}} - \underline{\text{change}} = \underline{\text{result}}$$

The Three Take From Subtypes

Table 2.1.9: Take From Problem Subtypes

Take From (Result Unknown)	Take From (Change Unknown)	Take From (Start Unknown)
Eight people are in a line. Five people leave the line. How many people are in the line now?	Eight people are in a line. Some people leave the line. Now there are three people in the line. How many people left the line?	Some people are in a line. Five people leave the line. Now there are three people in the line. How many people were in the line before?
$8 - 5 = ?$	$8 - ? = 3$	$? - 5 = 3$

The Seeds Problem

There were 15 seeds on the ground. A bird ate 7 of the seeds. How many seeds are on the ground now?

- a.  **Play: a student solves this problem** smarturl.it/cm3.2
- b. What type (and subtype) of addition or subtraction problem is this an example of?

The Animal Crackers Problem

Serenity had 17 animal crackers. Her brother ate some of them. Now Serenity has 15 animal crackers. How many animal crackers did Serenity's brother eat?

- a. **Face Partner A:** Use the cubes at your table to show your face partner how a child may use direct modeling to solve this problem.
Face Partner B: How else might a child go about solving this problem in a way that demonstrates making sense of the problem?
- b.  **Play: a student solves this problem** smarturl.it/cm3.4
- c. What type (and subtype) of addition or subtraction problem is this an example of?

Put Together and Take Apart Problems

These are also sometimes referred to as Part-Part-Whole problems.

Definition 2.1.12

A **put together or take apart problem** involves two sets that are either combined together or taken apart.

Put Together: $\underline{\text{part 1}} + \underline{\text{part 2}} = \underline{\text{whole}}$

Take Apart: $\underline{\text{whole}} - \underline{\text{part 1}} = \underline{\text{part 2}}$

There is no **change** in these problems, and there can be more than two parts.

The Put Together / Take Apart Subtypes

Table 2.1.14: Put Together / Take Apart Problem Subtypes

Put Together / Take Apart (Whole Unknown)	Put Together / Take Apart (Part Unknown)	Put Together / Take Apart (Both Parts Unknown)
There are three men and five women in a line. How many people are in the line?	Eight people are in a line. Three of the people are men and the rest are women. How many women are in the line?	Eight people are in a line. If they are all men or women, how many men and women could there be?
$3 + 5 = ?$	$3 + ? = 8$ $8 - ? = 3$	$? + ? = 8$

The Dogs and Cats Problem

Cynthia has 2 dogs and 3 cats. How many pets does Cynthia have?

- a. **Face Partner A:** Use the cubes at your table to show your face partner how a child may use direct modeling to solve this problem.
Face Partner B: How else might a child go about solving this problem in a way that demonstrates making sense of the problem?
- b. What type (and subtype) of addition or subtraction problem is this an example of?

Additive Compare Problems

Definition 2.1.17

An **additive compare problem** involves a comparison between the sizes of two sets, where one set is described as being larger (or smaller) than the other set by a certain amount.

$$\underline{\text{bigger}} - \underline{\text{smaller}} = \underline{\text{difference}}$$

$$\underline{\text{smaller}} + \underline{\text{difference}} = \underline{\text{bigger}}$$

$$\underline{\text{difference}} + \underline{\text{smaller}} = \underline{\text{bigger}}$$

The Additive Compare Subtypes

Table 2.1.19: Additive Compare Problem Subtypes (versions with “more”)

Additive Compare (Difference Unknown)	Additive Compare (Bigger Unknown)	Additive Compare (Smaller Unknown)
Eight people are in a line. Five people are in a second line. How many <i>more</i> people are in the first line than in the second line?	Five people are in a line. A second line has three <i>more</i> people in it than this line. How many people are in the second line?	Eight people are in a line, which is three <i>more</i> people than in a second line. How many people are in the second line?
$5 + ? = 8$	$5 + 3 = ?$	$8 - 3 = ?$
$8 - 5 = ?$	$3 + 5 = ?$	

Additive Compare, Continued

Table 2.1.19: Additive Compare Problem Subtypes (versions with “fewer”, continued)

Additive Compare (Difference Unknown)	Additive Compare (Bigger Unknown)	Additive Compare (Smaller Unknown)
Eight people are in a line. Five people are in a second line. How many <i>fewer</i> people are in the second line than in the first line?	Five people are in a line, which is three <i>fewer</i> people than in a second line. How many people are in the second line?	Eight people are in a line. A second line has three <i>fewer</i> people in it than this line. How many people are in the second line?
$5 + ? = 8$	$5 + 3 = ?$	$8 - 3 = ?$
$8 - 5 = ?$	$3 + 5 = ?$	

The Pencils Problem

Isabella has 4 pencils. Natalie has 9 pencils. How many more pencils does Natalie have than Isabella?

- a. **Face Partner A:** Use the cubes at your table to show your face partner how a child may use direct modeling to solve this problem.
Face Partner B: How else might a child go about solving this problem in a way that demonstrates making sense of the problem?
- b.  **Play: a student solves this problem** smarturl.it/cm3.5
- c. What type (and subtype) of addition or subtraction problem is this an example of?

Which Amount Is the Smaller One?

Example 2.1.20

There were four children in the pool who are not wearing goggles. The other six children in the pool are wearing goggles. How many more children are wearing goggles than are not wearing goggles?

Warning 2.1.21

One of the number facts associated with [Example 2.1.20](#) is $6 - 4 = 2$. The **smaller** amount in this problem is not 2. Although 2 is the smallest number that appears in this number fact, it does not represent the size of one of the two sets that we are comparing.